EFY TESTED
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MILLISECOND CLOCK DISPLAY Using ESP32 RTC

SOMNATH BERA

The author is fascinated by clocks and has made clocks of several types, but never with a millisecond display due to the huge processing power required for it. He was delighted when he could successfully run the 8.9cm (3.5-inch) ILI9488 TFT display in 8-bit mode with an ESP32 microcontroller. In 8-bit mode, it needs 12 GPIO pins for the display and it is very fast, and there lies the success of this millisecond display clock.

TABLE 1
BILL OF MATERIAL

Components	Quantity
ILI9488 TFT display (MOD3)	1
ESP32S (MOD1)	1
RTC DS3231 module (MOD2)	1
Micro USB cable	1
Coin cell	1

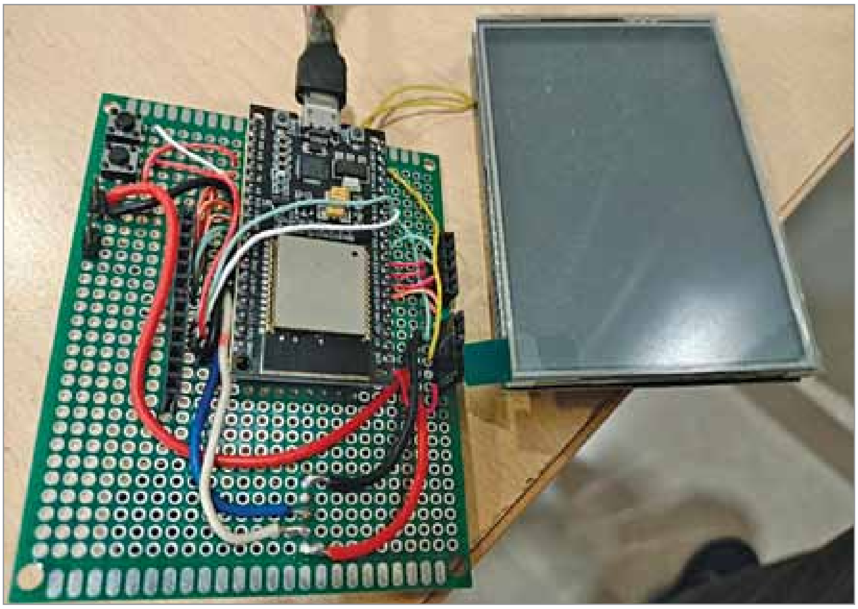


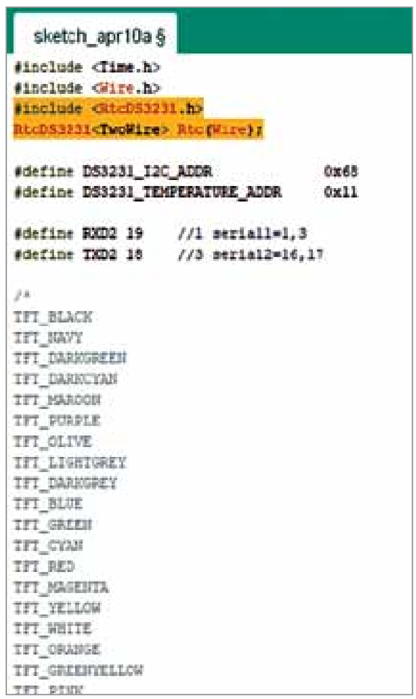
Fig. 1: Author's prototype

The 8.9cm TFT LCD touch screen display shield for Arduino Uno is available in the market. Although the TFT display fits very

TABLE 2
PIN CONNECTIONS OF ESP32 AND ILI9488 TFT DISPLAY

ILI9488 TFT Display Pin	ESP32S Pin
3.3V	3.3V
Gnd	Gnd
D0	7
D1	8
D2	1
D3	2
D4	3
D5	4
D6	5
D7	6
CS	15
DC	0 (keep empty)
RST	14
WR	17
RD	18

Fig. 2: TFT display used in the project



```
sketch_apr10a §
#include <Time.h>
#include <Wire.h>
#include <RtcDS3231.h>
RtcDS3231<TwoWire> Rtc(Wire);

#define DS3231_I2C_ADDR             0x68
#define DS3231_TEMPERATURE_ADDR     0x11

#define RXD2 19    //1 serial1=1,3
#define TXD2 18    //3 serial2=16,17

/*
TFT_BLACK
TFT_NAVY
TFT_DARKGREEN
TFT_DARKCYAN
TFT_MAROON
TFT_PURPLE
TFT_OLIVE
TFT_LIGHTGREY
TFT_DARKGREY
TFT_BLUE
TFT_GREEN
TFT_CYAN
TFT_RED
TFT_MAGENTA
TFT_YELLOW
TFT_WHITE
TFT_ORANGE
TFT_GREENYELLOW
TFT_PINK
```

Fig. 3: Code snippet